

INTRO TO PROGRAMMING, ATTRIBUTES, AND THE HIERARCHY OF A ROUTINE

In terms of training, the simplest definition of programming is *scheduling*. You are setting an order and time for planned workouts that will help you progress toward your goals. Thus, programming means having a plan for our workouts. The type of plan depends on the type of goals you have, as well as your current abilities. Previously, we discussed the mechanisms of strength and hypertrophy. To pull those concepts into focus, think of programming as how all of that information fits together on a large scale. This will be the information that is important when designing a workout program for you or your clients, if you are a trainer.

For example: You plan your exercise routine with repetitions at certain intensities, over certain numbers of sets, for a total volume of work. This volume of work is alternated or spread over a period of time, perhaps daily, weekly, biweekly, or monthly. In the case of elite athletes, the schedule of planning may be yearly, or on a quadrennial scale for Olympic athletes. Long-term planning allows your body to constantly progress. If you have competitions, you can target specific times that your body will be at its peak performance level.

The main concepts discussed in this chapter are the basics of programming and how they can be applied to bodyweight training.

INTRA-WORKOUT PROGRAMMING AND LINEAR PROGRESSION

Referring back to Chapter 1, we know that progressive overload is the key to consistent progression. Thus, the question is, “How do we implement progressive overload?” Let’s go back to the basic unit of programming, a single workout, and build from there.

The key of any single workout is to have a high enough amount of intensity and enough volume to stress the body so it will adapt to increase strength and hypertrophy. This is especially true if you are a beginner. Following the workout, it is important to rest for a certain amount of time, so that your body can recover and become stronger, more muscular, or both. There are multiple aspects of modifying workouts, which integrate the concepts that we learned about already—repetitions, sets, rests, tempo, intensity, volume, frequency, attributes, failure, work capacity, deloading, and plateaus. Of these, the most important factors in structuring a single workout are the intensity and volume.

- **Intensity** can be modified by increasing the difficulty of an exercise according to the amount of repetitions you can perform. This can be done through increasing the progression, utilizing a weight vest, or other methods.
- Volume can be modified in three different ways:
 - The amount of repetitions per set.
 - The amount of sets per exercise performed.
 - The total amount of exercises performed.

The body regulates itself through *homeostasis*, a concept that tells us there tends to be tightly maintained state in regard to body processes. When sufficient stress is placed on your body during exercise, it will adapt given sufficient recovery time. However, since the body has been forced to adapt to this change, it has already become somewhat resistant to the initial stress of the workout. Therefore, to continually grow stronger, it is illogical to repeat the same workout over and over again. What is the point of doing a workout repeatedly? If your body is already adapted to the stress, is repeating the same routine going to make you bigger or stronger? Surely not. Likewise, if you continually perform the same workout—or even perform a workout that causes adaptations but fail to progress after you have adapted—you may undertrain.

Undertraining is in most cases overhyped. As long as there is step-by-step progression—utilizing the concept of progressive overload—within workouts for your main exercises, then it is difficult to undertrain. However, there may be a point where there is not enough intensity or volume in a routine, which leads to undertraining. Undertraining may be the reason you are not progressing toward your goals, especially if you are only focusing on a few exercises.

One of the premier examples of intensity progression for beginners—linear progression via weights—is found in *Starting Strength* by Mark Rippetoe. This program works the core compound lifts—namely squats, deadlifts, power cleans, press, and bench press—for a fixed number of sets and repetitions in each workout. Linear progression increases the intensity of the exercises relative to the number of repetitions by adding weight. This process of adding weight to the lifts with each new workout—usually five or ten pounds per lift—forces constant adaptations with every workout, so that both strength and muscle mass can be drastically increased. This type of progressive overload is one of the most effective ways for beginners to progress.

- Monday: Squat 3x5 – 45 pounds
- Wednesday: Squat 3x5 – 55 pounds
- Friday: Squat 3x5 – 65 pounds
- Monday: Squat 3x5 – 75 pounds
- Wednesday: Squat 3x5 – 85 pounds
- Friday: Squat 3x5 – 95 pounds

For obvious reasons, with bodyweight exercises, you cannot effectively increase the intensity of an exercise by adjusting your weight, unless you have a weight vest or ankle weights. However, many people do not have access to these types of implements due to financial reasons. Beginner level adaptations will still occur quickly, but not quickly enough that one can move up a full progression level with every workout. For this reason, the best way to progress will be primarily through the aforementioned volume modifications:

1. Increasing repetitions.
2. Increasing sets.
3. Increasing the total amount of exercises performed.

If you have access to a weight vest or pulley system to use when exercising, it is often more effective to use those to make your bodyweight exercises more intense. This is because volume modification tends to be slightly slower with strength progression than intensity modification. Since your goal is primarily strength increases, you have constraints that you need to adhere to—such as not changing the speed of repetitions or changing the amount of rest between sets. Another constraint is to stick with exercises in the 5-15 repetition range if you are solely using weighted progressions. Additional constraints may be performing too many sets or exercises.

For beginners, there are a number of reasons why you should not increase exercises. First, more advanced bodyweight exercises are often vastly different than anything that most people have trained in before. Through these exercises the body is subjected to many movement patterns. Too many movement patterns make it difficult for the body to effectively learn. Imagine trying to learn ten songs at once on the piano, instead of just one or two at a time—it will not be very effective. Typically, you want to focus on learning a few at a time, so that you can practice them to efficiency before learning more.

Second, adding additional exercises often adds an additional two to three sets or more to a workout. While it is true that additional stress is needed to elicit continual adaptations, it is unlikely that you need that much additional overall volume. Adding the intensity of additional repetitions and sets by adding an additional exercise may be too much for your body to handle, which can lead to overuse injuries. For beginners, less is often more. An increase in one set or multiple repetitions is enough for beginners to progress well from one workout to the next.

Third, spreading out exercises between too many goals often leads to stagnation. Remember, you cannot optimally gain both strength and endurance at the same time, due to the adaptations that occur at the different ends of the repetition spectrum. Spreading yourself out between too many exercises and goals especially (which in the context of adding exercises is often the case) can lead to decreases in the overall quality of improvement in the areas that you are trying to improve.

STRESS, ADAPTATION, SUPERCOMPENSATION, FITNESS, AND FATIGUE

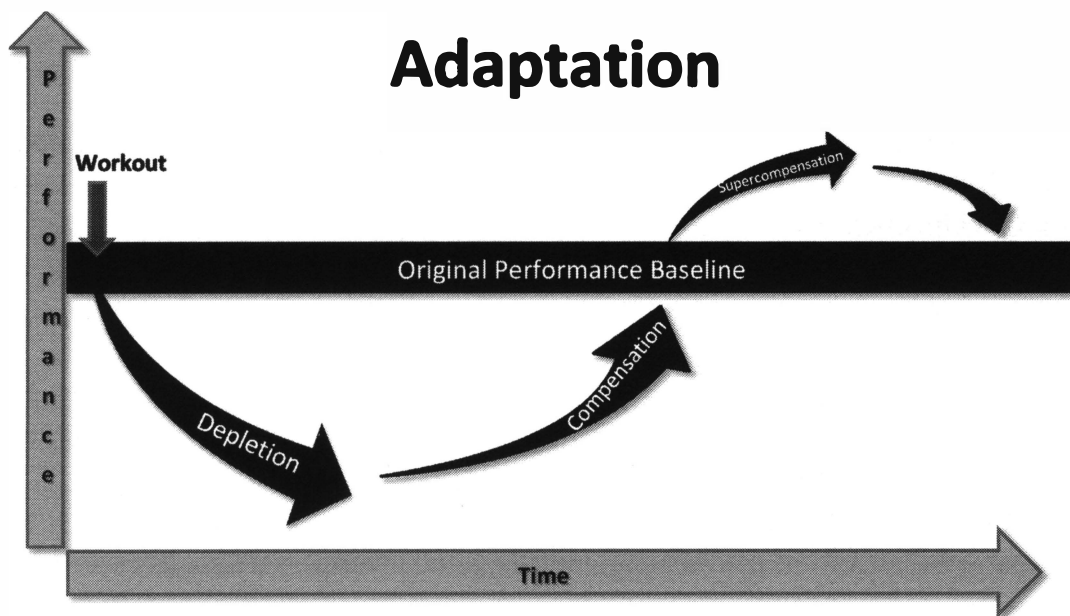
The first concept is understanding the nature of recovery. The easiest way to think of workouts versus recovery is the concept of the sympathetic nervous system versus the parasympathetic nervous system. In layman's terms, the sympathetic nervous system is termed the *fight or flight system*, whereas the parasympathetic nervous system is termed the *rest and digest system*. Workouts bring the nervous system into a state of arousal that is similar to fight or flight, where the body is being stressed in order to cause adaptations that will improve physiological attributes for subsequent training. On the other hand, the parasympathetic system is composed of all of the activities that promote recovery. These consist of, but are not limited to: sleep, diet and nutrition, de-stressing options (massage, saunas, and light activity like walking), mobility work, meditation, deep breathing, supplements, and the like.

Once you stop progressing from workout to workout, consider more advanced concepts if you want to increase your capacity to grow stronger. Thus, it is important to look beyond a single workout when planning

and structuring workouts. After a single workout, your capacity to perform additional work will decrease, typically within the next 24-48 hours. However, as your body heals with rest and proper nutrition, it will supercompensate, allowing it to come back stronger and better. *Supercompensation* is the concept that the combination of a few workouts that depress abilities more than normal will result in a rebound effect, which results in improvement that might not be seen from a single workout.

The training stimulus must pass a certain threshold to force good adaptations. This means you do enough to avoid undertraining, but not too much, which can actually cause so much damage that you do not gain any supercompensatory effect at all. Again, less is often more. Properly manipulating workouts, in order to avoid undertraining and overtraining, requires practice, time and observation.

Initially, a workout will produce both a positive effect, which is termed *fitness*, and a negative effect, which is termed *fatigue*. The latter results in depressed abilities. As you can see from the illustration, training too early after the first workout means that you will train again when you are under the initial baseline. This means that your abilities in the subsequent workout will be decreased compared to the first one. Training within this period of decreased abilities will result in a greater level of fatigue. This fitness increase is immediate in terms of strength, just like you can improve in other neurological based abilities such as balance within the same session; however, fatigue masks this increase in fitness. In terms of muscular hypertrophy, the adaptations occur slightly slower over time, because protein synthesis, which adds muscle mass, takes place over a 48-72 hour period following a workout.



On some level, once you increase the frequency or overall volume load of your workouts beyond a certain point, you will reach a stage at which your fatigue has not yet dissipated before you perform your next workout. This will often reoccur with the workouts that follow. An interesting thing is that you can still see progress—even significant progress—in strength and/or muscle gain, even with fatigue accumulating. However, it is inevitable that you will eventually “stall out,” resulting in a plateau. Accumulation of fatigue that is equal to your gains equals reaching a plateau in your training. The point at which gains stall is a good

measurement of when fatigue has overtaken fitness or supercompensatory adaptations, which allows continuous progress in spite of fatigue.

To counteract the fatigue buildup, you should do exactly what you would expect to do when fatigued—rest. A rest period between workouts for a beginner will typically be one day. Rest comes in many forms, ranging from taking a break from training, to working on skill work, to simply deloading total intensity, repetitions, sets, and the like. Fitness and fatigue are an alternative visualization of the traditional model, and are termed in “dual factor” theory. Let us discuss other factors involved before we can see why it makes a difference.

For beginners, 48-72 hours is the optimal time for recovery after a workout if you want to see enough supercompensatory effects that your performance is greater in your next workout. As previously mentioned, many beginner programs are initially based on the three times per week model, which allows 48 hour between two of the workouts and a 72-hour break after the third.

This, however, is not the fastest way to gain strength. If you recall, strength has two components—neurological adaptations and muscular adaptations. The body can build up a resistance to stress from both of these by repeatedly forcing it to adapt with workouts. This is why professional athletes often workout nearly every day of the week—sometimes multiple times a day—without getting sore. At the highest levels, the frequency of workouts will matter the most in gaining strength. You may have heard it said that strength is a skill. Repeating a certain movement over and over with progressive overload will lead to strength gains, which is why doing too many different exercises should be avoided if you want to gain strength.

The body has an abnormal capacity for increasing neurological strength, even at certain muscle sizes. For example, Olympic weightlifters from the 69 kg. (151.8 lbs.) weight class can put up amazingly heavy weights. The current world records in this weight class for the snatch is 165 kg (363 lbs.), and the clean and jerk is an impressive 198 kg. (435.6 lbs.). These movements entail taking a barbell that weighs the amounts listed from ground to overhead in one and two movements, respectively. This shows us that it is not necessary to be heavy to be incredibly strong.

The greatest capacity for strength gains comes from increases in neurological adaptations, but it is also the neurological system that adapts much slower relative to its potential compared to muscle mass. If you remember the pool analogy, the nervous system’s “pool” will tend to fill up slower than the muscles. The reason for this is that you can literally “force” the body to put on muscle mass if you lift heavy weights and eat like there is a famine coming. To overcome this neurological bottleneck, it is often advisable to train more than three times a week. For more advanced athletes, this may entail a few of these sessions being lower intensity, as opposed to pushing yourself every time. This will allow you to improve in your abilities more rapidly, but you may be more likely to burn out if the intensity of your workouts is too high and not balanced with decreases in volume.

This interplay between fitness and fatigue when training for strength is the reason why periodization is necessary after you reach intermediate and advanced levels.

BASIC PERIODIZATION AND INTER-WORKOUT STRUCTURE

The concept of *periodization* was pioneered by the Soviet Union scientists in the 1950-70s. Their goal was to dominate athletics, and this was the method they chose in which to do so. Periodization in sports is a way to

organize training by progressively alternating various aspects such as frequency, intensity, volume, repetitions, etc., in order to increase physical capacity. It is a plan that encompasses multiple workouts over a set amount of time. In typical Soviet periodization structure there are three components: the microcycle, the mesocycle, and the macrocycle.

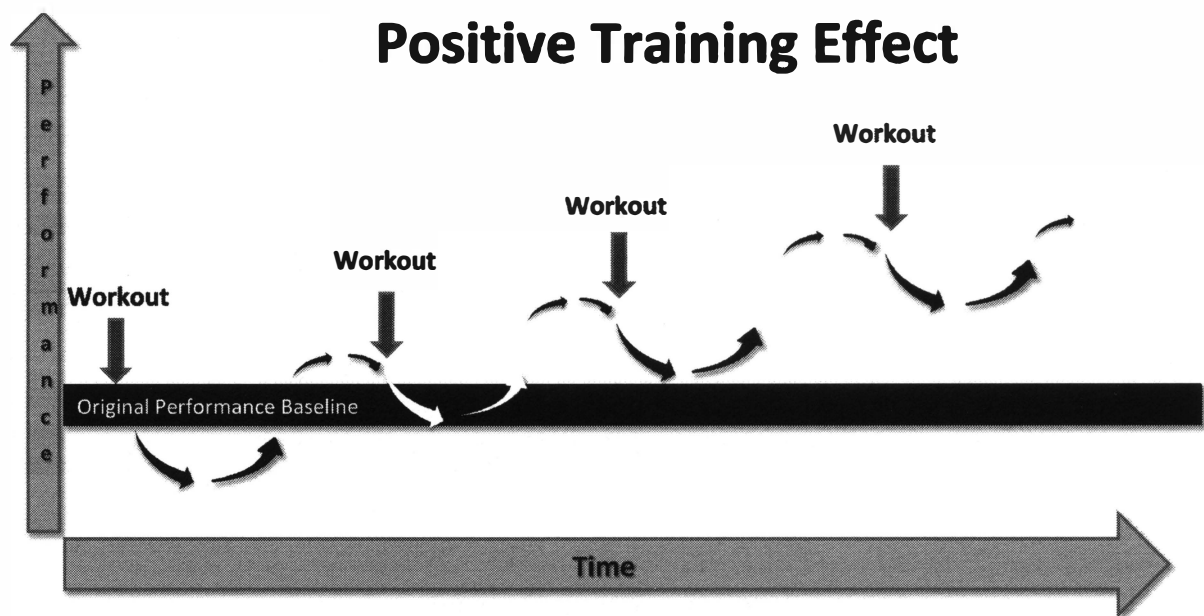
The microcycle is typically based around one week of training. It is generally organized into a specific attribute that one desires to focus on. In the old Soviet model there were generally four microcycles that were used: a preparatory phase, hypertrophy phase, strength phase, and a power phase.

A mesocycle is a combination of four to eight microcycles. The amount of microcycles depends on how many phases there are in a program. So in the context of the above types of microcycles, in the archetypal Soviet system there was generally one to two preparatory phase microcycles, one hypertrophy microcycle, one strength microcycle, and one power microcycle. These microcycles were often followed by a deload or rest week, in order to dissipate fatigue. Therefore, a typical mesocycle was six weeks long.

A macrocycle is simply a combination of mesocycles, which is often planned so that an athlete will finish their final mesocycle close to their competition date. Because an athlete will still be training hard toward the date of their competition, they will be strong, but this approach will allow enough rest to dissipate excess fatigue, thus allowing them to peak during the competition so they can lift weights well above their previous personal records, hopefully allowing them to win their competition. One “macrocycle” that covers 24 weeks may look like this:

Mesocycle	Microcycle – 1 Week (Preparatory)
	Microcycle – 1 Week (Preparatory)
	Microcycle – 1 Week (Hypertrophy)
	Microcycle – 1 Week (Hypertrophy)
	Microcycle – 1 Week (Strength)
	Deload / Rest Week
Mesocycle	Microcycle – 1 Week (Preparatory)
	Microcycle – 1 Week (Hypertrophy)
	Microcycle – 1 Week (Hypertrophy)
	Microcycle – 1 Week (Strength)
	Microcycle – 1 Week (Strength)
	Deload / Rest Week
Mesocycle	Microcycle – 1 Week (Preparatory)
	Microcycle – 1 Week (Hypertrophy)
	Microcycle – 1 Week (Strength)
	Microcycle – 1 Week (Strength)
	Microcycle – 1 Week (Power)
	Deload / Rest Week

	Microcycle – 1 Week (Preparatory)
	Microcycle – 1 Week (Hypertrophy)
Mesocycle	Microcycle – 1 Week (Strength)
	Microcycle – 1 Week (Power)
	Microcycle – 1 Week (Power)
	Deload / Rest Week



Notice how there may be a different emphasis in each mesocycle, depending on your particular goals. In particular, the first mesocycle in this example focuses more on preparation, hypertrophy, and strength, while the latter mesocycles ultimately focus more on strength and power. The different phases mean variations in the intensity of the exercise. In terms of the specific “attributes” within a cycle, the typical Soviet structure was described like this:

- A preparatory phase tends to be a slow increase in volume and intensity coming up from an off week. It was basically a “ramp in” for a serious workout directly after a deload week.
- A hypertrophy phase tends to range in about 60-85% 1 RM (about 5-12 repetitions).
- A strength phase tends to range around about 80-100% 1 RM (about 1-8 repetitions).
- A power phase tends to utilize lighter weights around 40-60% 1 RM (about 3-6 repetitions) and focuses on pure acceleration.

There are other types of workouts that focus on specific attributes, such as *Dynamic Effort* from Westside Barbell. Dynamic Effort (DE) is also termed *speed work*, where 40-60% 1 RM is used with additional accommodating resistance such as chains, bands, or other implements with about 25-30% 1 RM. Speed work is

about utilizing lighter weights and focusing on accelerating the bar as quickly as possible. The accommodating resistance is used to work on sticking points or easier points in the lift.

How does this structure apply to us? It is extremely convenient for most people to operate on the weekly microcycle schedule because the vast majority of occupations around the world are based on a weekly schedule. Thus, any intermediate or advanced programming will generally be for the seven-day week. Secondly, microcycles teach the valuable lesson that you must consider overall volume on a weekly basis, rather than workout to workout. Remember, as you grow stronger and more muscular, your body becomes more resistant to stress, which makes it more difficult to force the adaptations you desire. Thus, your focus must shift from a workout to workout structure to a weekly model. It may seem counterintuitive, until you realize that it is easy to plateau for long periods of time once you get past the beginner strength level. You probably know someone who has been going to the gym for years, yet is still using the same weights or even looks the same. This person has not yet learned how to structure their workouts to implement progressive overload, which would allow them to continue making gains.

When you become sufficiently strong, your workout schedule may switch to biweekly or monthly scheduling. Pro athletes may get to the point where they utilize yearly scheduling, while Olympic athletes may get to the point of quadrennial scheduling. This is what it means to *increase the complexity of programming*, which takes place as you become stronger and move into the intermediate and advanced ranges. This takes years of consistent work, so the average person who is beginning bodyweight training will not need to worry about it for awhile. Instead, mesocycles will be the primary part of a beginner program. Generally speaking, the mesocycles should be anywhere from four to eight weeks long, depending on when you plateau or feel fatigue sufficiently catching up with you. These mesocycles will be long enough that your body will adapt and become stronger. After you complete a mesocycle, you will often have a rest week, in order to dissipate additional fatigue, allow any aches and pains to go away, plan prehabilitation or rehabilitation to recover fully, evaluate your goals, plan or program, and rest for your next mesocycle.

Rest weeks are extremely critical to your progress. If you have never properly utilized rest weeks, you likely have not progressed very far in your training unless you intuitively know how to train effectively. A safe estimate would be that only ten percent of those who get into bodyweight training know how to do this. Thus, the vast majority of athletes must learn the concept of rest or deload weeks if they desire to make long-term progress.

There are entire books written on periodization. The material covered here is just a quick summary. There are three particularly recommendable books. *Supertraining* by Mel Siff is a very dense read. Two other solid titles are *Science and Practice of Strength Training* by Vladimir Zatsiorsky and *Periodization: Theory and Methodology of Training* by Tudor Bompa.

ATTRIBUTE TRAINING

The notion of “phases” of microcycles is important because they are geared to develop specific attributes. This applies not just for preparation, hypertrophy, strength, power, and endurance, but also to other facets of training as well. Thus, it is important to recognize that routines are made up of multiple parts that all work in conjunction with one another.

Here are the primary questions that this section will address when it comes to constructing a routine:

- Can I do this exercise every day?
- Can I do this exercise on my off days?
- Can I do this exercise with my workouts?

To better understand programming a routine, you must first understand how the different concepts that make up a routine affect your ability to execute that routine. There should be a purpose to every exercise you put into your routine, and that purpose can be categorized into working specific attributes.

For example, a short list of the various attributes that may be trained with exercises in a routine are as follows: strength, hypertrophy, endurance, cardiovascular, mobility, flexibility, stability, skill, endurance, etc. There are more than this, and they can be categorized in a number of ways. Jim Cawley and Bruce Evans of *Dynamax* originally categorized ten physical areas of development that the now ubiquitous CrossFit adopted before it became extremely popular. These are cardiovascular/respiratory endurance, stamina, strength, flexibility, power, speed, coordination, agility, balance, and accuracy.

This book focuses on constructing a routine dedicated to specific increases in strength and hypertrophy. Typical exercises that are used for strength and hypertrophy are any of the progressions that focus on improving strength, such as isometrics (planche, front lever, back lever) and full range of motion exercises (pushups, dips, handstand pushups, rows, pull-ups).

The three primary factors in a workout are strength, hypertrophy, and endurance. The main thing that separates these attributes are the intensity of the repetition in sets and thus how many repetitions of the exercise you perform in a single set.

- **Strength** – Primarily developed in the 1-8 RM range, best executed with close-to-failure (but not-to-failure training) and 3-minute rest times.
- **Hypertrophy** – Primarily developed in the 5-15+ RM range, best executed with failure training and anywhere from 1-3 minute rest times.
- **Endurance** – Primarily developed in the 15-20+ RM range, best executed with failure training and 30 to 90 second rest times.

Most of the other attributes significantly vary in their training. For example, with cardiovascular-specific training the attributes that are trained depend on the intensity of the exercises. The intensity almost always determines the frequency for most non-workout specific attributes. Let's use running as an example, but this also applies for biking, swimming, and other endurance related activities. For example, these are the cardiovascular factors that you may program after strength or hypertrophy work in a workout:

- **Recovery** – If it is light work where you are running simply to get blood flowing and you end a session breaking a slight sweat and feeling better than you started, this can be performed every day.
- **Moderate** – If you are running for moderate intensity cardiovascular work that is not necessarily aimed at training cardiovascular attributes (but instead for general health), it can be added to a rest day.
- **High or Workout-Focused** – If you are actually training to improve your cardiovascular capacity in order to run farther, faster, or longer for a race or competition, this type of training may be its own workout and may replace workouts in your routine geared toward strength, hypertrophy, or endurance.

The main thing to understand here is that almost any type of high-intensity work where you are training specific attributes is only going to detract from your recovery and thus impair other portions of your routines. Thus, the answer to the question “*Can I do this exercise every day?*” will vary. Here are a few more examples to help explain.

Flexibility is a bit of an odd ball in that it can technically be trained everyday, but it can also interfere with your routine. This is because flexibility is an eccentric exercise. As a muscle elongates in the body the passive tension will ramp up in the muscle as it is contracted by the nervous system. This is so the muscle will not move into a dangerous position. Thus, flexibility can be trained frequently, but you have to factor it in to the overall context of your routine. If you have many leg-focused exercises, it may not be a good idea to train many flexibility exercises with the legs, as the soreness or fatigue caused by stretching may impair your ability to perform optimally.

Mobility is one of the attributes that can be trained every day. It involves taking a muscle through its range of motion, either actively or passively, and it is not geared toward improving any specific attribute. If you are looking for optimal health, doing mobility every day will help maintain a good range of motion and ensure that your joints will work correctly and not stiffen up. This is important for older adults as loss of mobility and strength—especially as you age—is correlated to increased mortality.

Skill Work depends on the intensity of the skills. As stated before, handstands may be strength work for beginners at first and thus may be a part of the actual workout. However, once balance becomes the priority of training and strength becomes less of an issue, it can be trained more frequently. Skill work that is focused on balance can be trained every day. Isometrics are not always skill work as planche, back lever, and front lever are never about balance, but training strength.

We are not going to discuss other exercise attributes such as stability on an in-depth level. You should have a strong sense of the approach from the previous examples, but it is up to you to draw conclusions for yourself. Developing critical thinking skills in terms of programming and planning routines is going to be important to learn how to do it effectively. To aid in categorization and classification of specific exercises that train attributes, think of them in the following manner:

- **Workout Factors** – These are generally performed two to four times per week because there needs to be significant recovery between each session due to high intensity, training to failure, or other high volume work. Variation beyond this frequency on a weekly basis is typically reserved for those who have significant experience designing exercise programs.
- **Recovery and Optimization Factors** – These can be performed three to infinite times per week. These factors are aimed at facilitating recovery of the body through movement, soft tissue work, or low-intensity exercise that gets the blood flowing. Most of what falls into this category should leave you feeling significantly better than when you started, and will not impair workout sessions performed later in the day or the day after. This includes lighter-intensity flexibility work, recovery jogs and walks, mobility work, light skill work, and the like.
- **Rehabilitation Factors** – Rehabilitation and prehabilitation factors are specific. The main thing to understand about these is that rehabilitation is not always focused on increasing strength, though that can be a part of it. Typically, when a tissue is injured you will want to go through a phase where you begin with low-intensity workouts and build up to higher intensity workouts. High repetitions at low intensity are typically performed after a routine, where you work on

prehabilitation or rehabilitation of the specific body parts that need extra work. However, these can also be performed on off days.

This was not emphasized in the first edition of this book, but the primary concept to take away from this section is “everything depends.” The type of template that you select for a routine does not limit you in the amount of goals that you want to set for training different attributes. In terms of overall goals, the type of template you pick for a routine will only matter when it comes to the main attributes you are working to develop. These attributes are strength, hypertrophy, and/or endurance. Skill work, flexibility, mobility, prehabilitation, rehabilitation, and the like often are easier to train with a routine. However, they can be trained on your off days, which gives you the flexibility to move things around to fit your schedule. Many athletes and coaches feel limited in what they can accomplish when given a certain type of template for a routine that does not allow this much flexibility.

THE BASIC HIERARCHY OF A ROUTINE

When you are constructing a routine, it is important to follow a few simple rules that will maximize the amount of training “ability” you have in a single workout.

If you have a series of tasks to complete, what order would you put them in to maximize your abilities and increase performance for subsequent workouts?

This section aims to answer that question. Let’s outline a general structure of a full-body routine that is performed three times per week. Then we will focus on building a routine on this foundation.

1. Warm-Up and Mobility
2. Skill or Technique Work (Handstands, Flips, Gymnastics Tumbling, Breakdancing Work, etc.)
3. Power, Isometrics, Eccentrics, Regular Strength Work
4. Endurance, Metabolic Conditioning, Tabata Method, Interval Training, Specific Exercises, etc.
5. Prehabilitation, Flexibility Work, Cool Down

These are the five main categories that you should construct a workout routine around. The most detailed section will be about strength training since this book is primarily about learning how to become stronger; however, we will also discuss extensively how all of these other categories will interact with the strength category and how they can also be used in the context of your sport (if you have one). This will allow you to learn these principles, implement them into your training, and broadly apply strength and conditioning principles to your athletes if you are a coach. Remember, not all of the items above must be in your routine, nor must they all be performed on the same day.

The best introduction to learning how to construct a routine is through examples. So, here is an example of a basic beginner routine of about Level 3-4 on the bodyweight progression charts. This example is on how to construct a workout routine based on what is effective. It may include concepts from other bodyweight sources, including but not limited to: my gymnastic coaches, Roger Harrell, Blair Lowe, Jim Bathurst, Ido Portal, and others. If you are an avid Reddit user you may realize that this is similar to the */r/bodyweightfitness* beginner routine. Phi helped fill in the content, and I played a role in structure and routine construction. Shout out to all of the mods and contributors, which include: *phi*, *phraktur*, *SM*, *m092*, *iscg*, *antranik*, *kay-etch*, and *Solfire*.

Let's use gymnastics for this example because it is the discipline out of which this book is born. However, these concepts have helped thousands of athletes from parkour/free running, climbing, break dancing, martial arts, swimming, and other bodyweight and non-bodyweight sports.

Goals

10 Freestanding Handstands
 5 Strict Muscle-ups
 5 Planches
 5 Front Levers
 10 Pistols
 10 Vertical V-Sits
 5 Iron Cross's

(These are arbitrary goals selected as examples.)

Warm-Up

1. Blood Flow: 10-20 Burpees, 60s Crawling (or variable distance, such as 100 meters)
 2. Mobility: 15x Wrist Circles, Shoulder Circles, Bodyweight Squats, and any other body part or joint that needs to be warmed up, 60s of Support Hold Work (parallel bars or rings or chairs/counters), 5x Skin the Cat / German Hangs
 3. Positional Drills: 30-60s of Plank, Both Side Planks, Reverse Plank, Hollow, and Arch Holds.
- (Organized according to: classification of exercise, amount of repetitions, type of exercise.)*

Skill Work

5-10 minutes of Handstand Work, performed against the wall.

Strength Work

Pull-ups: 3x5→15 with 3 minutes of rest at 10x0 tempo
 Dips: 3x5→15 with 3 minutes of rest at 10x0 tempo
 Wide Ring Rows: 3x5→15 with 3 minutes of rest at 10x0 tempo
 Rings Pushups: 3x5→15 with 3 minutes of rest at 10x0 tempo
 Squats (pistol progression or barbell): 3x5→15 with 3 minutes of rest at 10x0 tempo
 Deep Step-ups: 3x5→15 with 3 minutes of rest at 10x0 tempo
 L-sit for 60s total, in as many sets as needed, not to failure
 Compression Work for 3x10s

(Organized according to: exercise, exercise order, sets x reps with progressive principle, rest time, tempo.)

Prehabilitation, Isolation Work, Flexibility Work, and Cool Down

3x1 minute sets of Rice Bucket for the wrists
 3x10 Biceps Curls
 3-5x30s Splits Holds
 3-5x30s German Hangs
 3-5x20s Back Bridges
 1 minute of Deep Breathing (in through the nose, out through the mouth)

STOP READING. TAKE ACTION.

1. Write down any skill work you want to learn. Also, start to think of the types of exercises you desire in your routine and how you would order them based on the information above.
2. Write down any previous injuries you have had or problematic joints, muscles, or tendons that have or may have been bothering you in the past. Also, write down what you have been doing for rehabilitation or prehabilitation to solve these injuries.
3. We will take into account how to integrate all of these factors into a routine and how to avoid any contraindicated movements while continually working toward our goals.
4. Assess your soft tissue structures by using your hands to massage in and check for tightness, trigger points, or other restrictions or chronic problems in your musculature, tendons, or around the joints. Write these down too.
5. In addition, check and evaluate your flexibility and mobility. The main ones we want to check at this point include your wrists, elbows, shoulders, the straddle stretch, and pike stretch.

Next, use your list of refined goals in conjunction with the skill and progression charts. Select exercises that correspond to your goals and take the following into consideration.

- Any of the two to four skill-based exercises that you are working on are at least one or more levels below your competency level, so it would be optimal to practice them without significant fatigue.
- Any of the pushing and pulling goals have exercises selected to progress toward them. Generally, one to two goals with a total of two to three exercises planned for them.
- The exercises you select should work around any injured body parts. For example, you should select exercises that do not aggravate the existing injuries. Likewise, if you are working around any injured areas you should include prehabilitation or rehabilitation work in order to facilitate the healing process so that it does not linger.
- Since a high degree of flexibility is required to optimally progress with many upper-level body-weight strength skills, it is likely that a lot of hip and shoulder flexibility work will be required for those with non-flexibility athletic backgrounds.
- Group all the warm-up, skill development, and mobility exercises into the beginning, strength work (strength/power) into the middle, and end with cool down, flexibility, and rehab.
- Now, order all of these exercises according to your goals or specific areas you want to work on.

CHAPTER 5 SUMMARY

INTRODUCTION TO PROGRAMMING, ATTRIBUTES, AND THE HIERARCHY OF A ROUTINE

Knowledge Base

Programming is the concept of changing workout routines via intensity, volume, and frequency, which results in progressive overload. For beginners this is often performed by increasing the repetitions, sets, or the amount of exercises performed.

Once you become more advanced it is important to realize that things are going on underneath the surface of a training program. This is why understanding the macro effects of stress and adaptation in the context of weekly programs is important, especially knowing that a program affects fitness and fatigue, which can allow one to plan gains through supercompensation.

Attribute training is the concept of understanding how the parts of a routine work, so you can adjust them on the fly if needed. For example, aside from strength and hypertrophy work, there may also be flexibility, mobility, skill work, and other elements that fit into a workout routine. Attribute training lets you know that some of these components do not necessarily have to be solely in a workout routine, but can be performed many times throughout a week.

Finally, the basic components of a routine are as follows:

- Goals
- Warm-Up – Blood Flow, Mobility, Position Drills
- Skill Development – The development of sport-specific skills or handstands
- Strength Work – Followed by other work, such as endurance or other cross training
- Prehabilitation, Isolation Work, Flexibility Work, and Cool Down

The warm-up tends to consist of lower-intensity exercises. The goal of this is to raise core temperature and get the nervous system and muscles operating at full capacity. In addition, some degree of prehabilitation, mobility work, or stretching may be integrated if it benefits the ensuing workout.

Skill development can be integrated into the warm up if time is a factor. This tends to focus on lower-intensity skills that need a lot of practice to attain.

The strength work of the workout is the power, eccentrics, isometrics, and dynamic movements. These make up the core of the work you will be doing to move toward your goals. This may be followed with other work as necessary.

The cool down period is composed of rehabilitation or prehabilitation work, as well as isolation (to improve specific injury conditions or to stave off ones from development). Likewise, mobility and flexibility work is often better integrated into a routine at this point, as the body is more responsive after the nervous system and muscles are tired from working out.

Application

Look at your categorized goals and use this template for building your workout routine. In this next part, we will go over how to assemble the specific components into the routine you will perform during your first workout.